

Comparing Distributions:
Kolmogorov-Smirnov (KS) Test

PHYS 4730
Fall 2026
Lab 16

Q: Are two RV distributions the same?

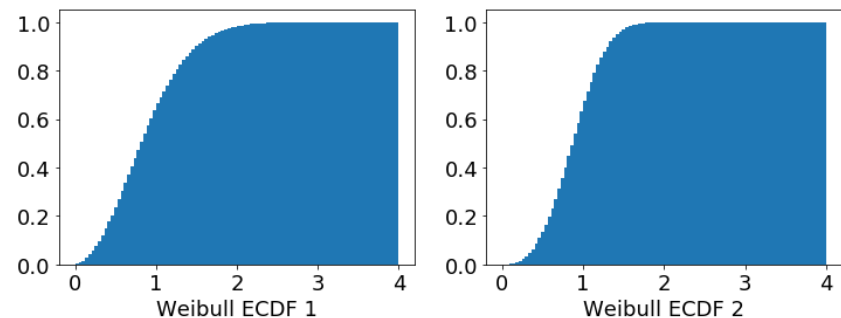
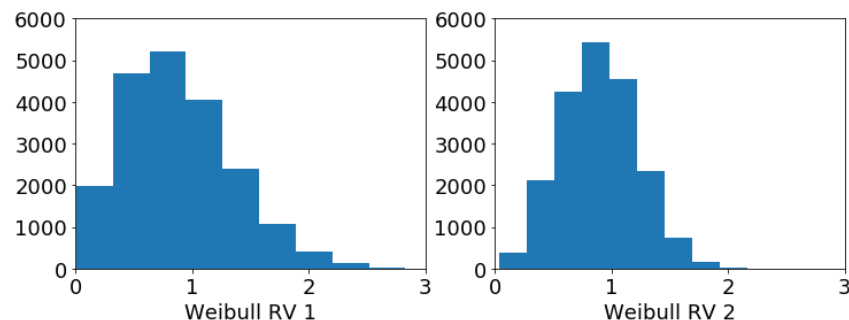
- Compare PDF and EPDF – “one-sample”; How likely is it that we’d see a set of samples like this, if drawn from the given probability distribution?
- Compare EPDF and EPDF – “two-sample”; How likely is it that we’d see two sets like this, if drawn from the same (unknown) probability distribution?

Example: KS_test_etc.ipynb

```
rvvec1 = rnd.weibull(Shape1,(N,1)).flatten()  
rvvec2 = rnd.weibull(Shape2,(N,1)).flatten()
```

```
### Find ECDFs for two datasets ###
```

```
nbins = 100  
aa = 0.0  
bb = 4.0  
  
dx = (bb-aa)/nbins  
  
ecdfX = np.arange(aa,bb,dx)  
  
plt.figure(figsize=(12,4))  
  
plt.subplot(1,2,1)  
ecdfY1,b,p = plt.hist(rvvec1, bins=nbins, range=[aa,bb], density=True, cumulative=True)  
plt.xlabel("Weibull ECDF 1")  
  
plt.subplot(1,2,2)  
ecdfY2,b,p = plt.hist(rvvec2, bins=nbins, range=[aa,bb], density=True, cumulative=True)  
plt.xlabel("Weibull ECDF 2")
```



Example: KS_test_etc.ipynb

The KS-test strategy is to look at the *cumulative distributions* and quantify their separation. Formally, for the “one-sample” case:

$$D_n = \sup_x |F_n(x) - F(x)|$$

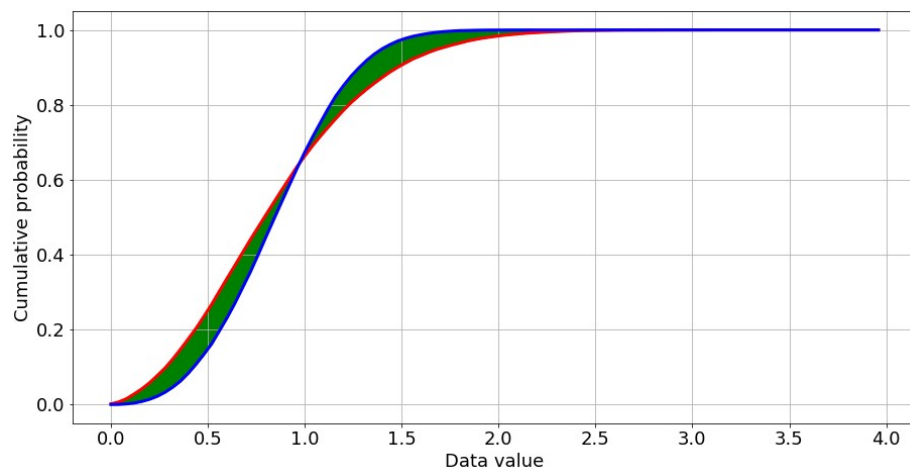
where $\sup_x$ is the “supremum” of the set of distances.

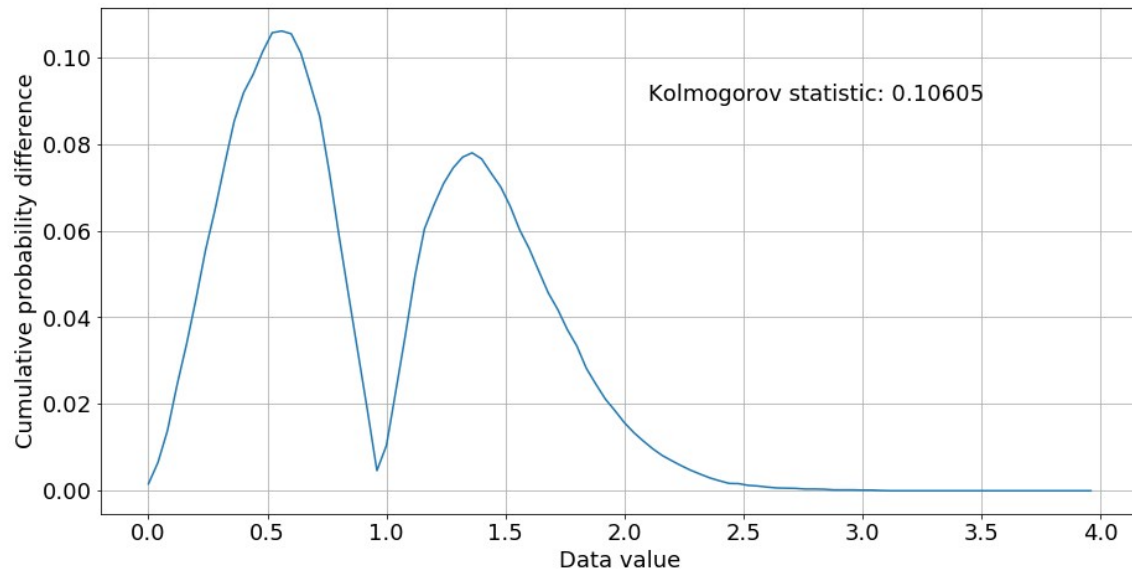
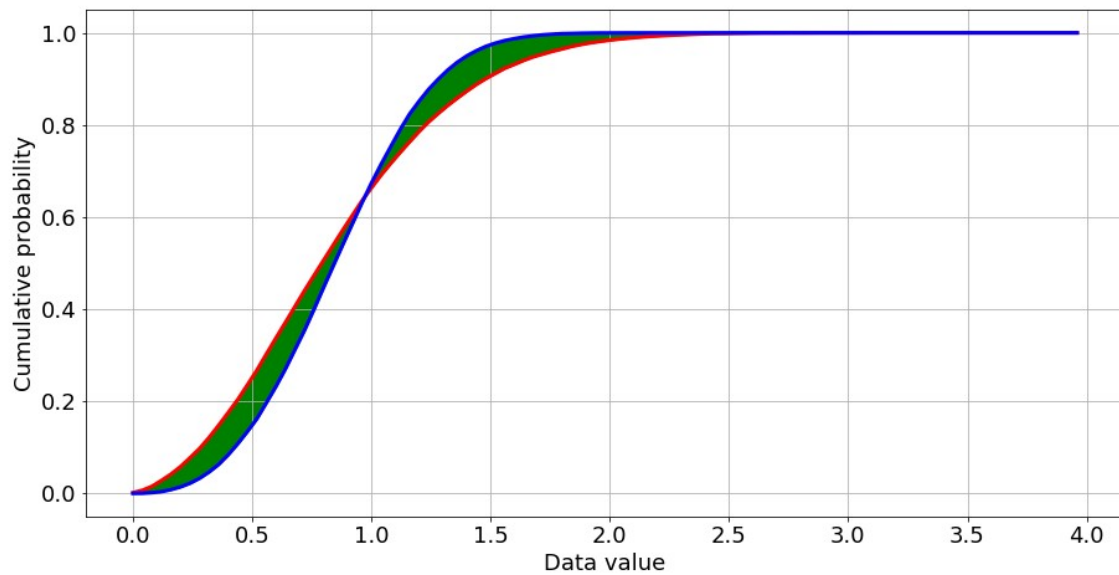
For the “two-sample” case:

$$D_{n,m} = \sup_x |F_{1,n}(x) - F_{2,m}(x)|$$

If the RV is from the given CDF (one-sample) or if the two RVs are from the same (unspecified) CDF (two-sample), then the expectation is that

$$D_{n(m)} \longrightarrow 0 \text{ as } n(m) \longrightarrow \infty$$





Determining p -value from Supremum

Now, how do we use the supremum D to determine the chance probability that the distributions are the same?

- a) One-sided (PDF vs EPDF) comparison with data size n : Kolmogorov (1933) showed that in the limit of large n , the probability $P(\sqrt{n}D_n \leq \lambda)$ will converge to a CDF given by the Kolmogorov distribution

$$P(\sqrt{n}D_n \leq \lambda) \rightarrow 1 - 2 \sum_{k=1}^{\infty} (-1)^{(k-1)} e^{-2k^2 \lambda^2}$$

thus the p -value will be given by

$$p = P(D_n \geq d_{obs}) = 1 - P(D_n \leq d_{obs}) = 2 \sum_{k=1}^{\infty} (-1)^{(k-1)} e^{-2k^2 n d_{obs}^2}$$

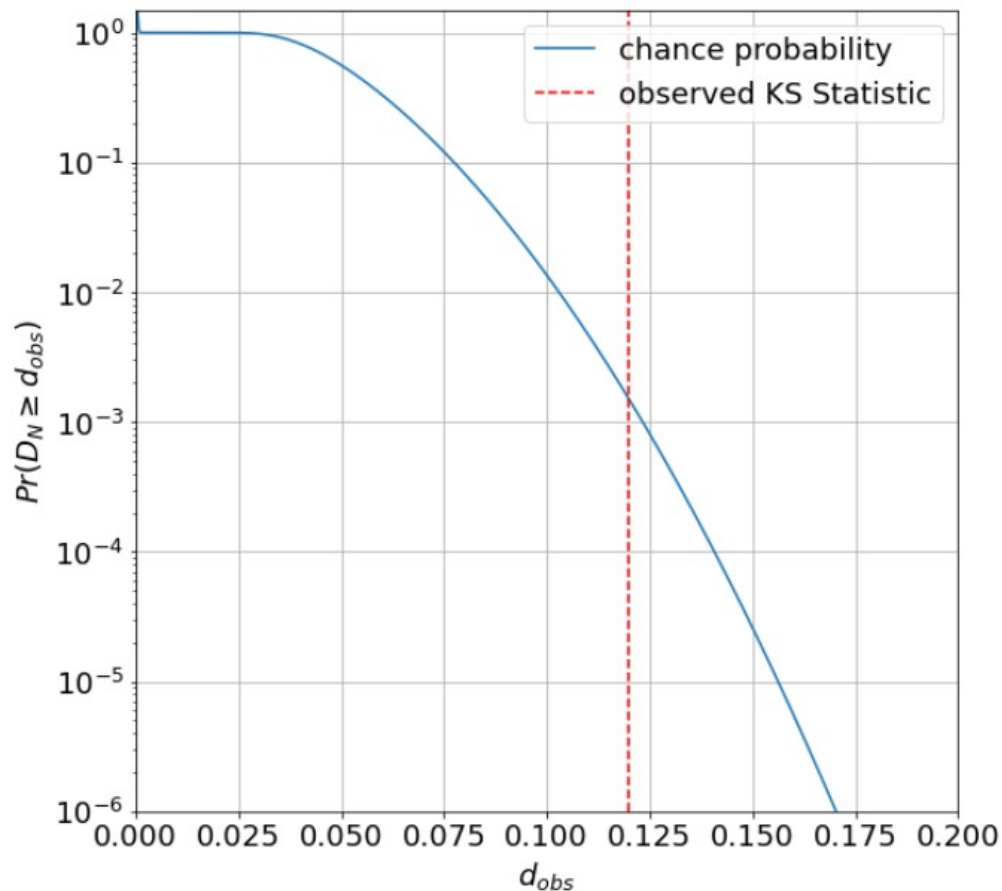
- b) Two-sided (EPDF vs EPDF) comparison with data sizes n, m : Define the effective sample size by

$$N = \frac{nm}{n+m}$$

then

$$p = P(D_N \geq d_{obs}) = 2 \sum_{k=1}^{\infty} (-1)^{(k-1)} e^{-2k^2 N d_{obs}^2}$$

Determining p -value from Supremum



Interpret p-value in terms of the *Null Hypothesis*:

Assuming the (E)PDFs are drawn from the same parent distribution, the p-value is the fraction of similar datasets for which we'd expect to obtain the same or greater value of the supremum $D_{n(m)}$.

Today's Lab

```
### So, that's what the Kolmogorov statistic is. ###
### Now, scipy.stats will do all of this for us, ###
### right up to computing p-value. ###
###
### There are a couple of ways of doing this. ###
### Compare scipy results to our answer. ###

output1 = SciStat.kstest(rvvec1,rvvec2,alternative='two_sided')
output2 = SciStat.ks_2samp(rvvec1,rvvec2)

print("")
print("Scipy Method 1:")
print("The Kolmogorov statistic is:      %7.5f" % output1.statistic)
print("The p-value is:                  %7.5e" % output1.pvalue)
print("")
print("Scipy Method 2:")
print("The Kolmogorov statistic is:      %7.5f" % output2.statistic)
print("The p-value is:                  %7.5e" % output2.pvalue)
print("")
print("Our method:")
print("The Kolmogorov statistic is:      %7.5f" % kstat)
print("The p-value is:                  %7.5e" % mypval)
print("")
```

- Apply KS test to separation of Gaussians.
- (No need to calculate supremum, etc. yourself. Use scipy.stats.)